YILDIZ TECHNICAL UNIVERSITY
FACULTY OF ELECTRICAL AND ELECTRONICS
DEPARTMENT OF CONTROL AND AUTOMATION ENGINEERING

QUADROTOR MODELLING CONTROL AND IMPLEMENTATION ON REAL SYSTEM

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ABSTRACT

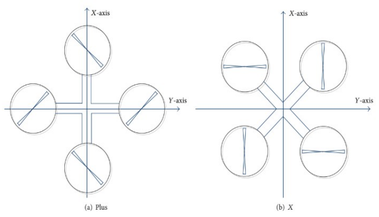
This study focuses on the mathematical modeling of a quadrotor system, the design of a PID controller, and the implementation of a real hardware system. The original contribution of this project lies in encompassing the mathematical modeling of the quadrotor system, the design of a P-PID controller, and the implementation of a real hardware system, representing a significant step forward in drone technology. The innovation of this work is in offering a comprehensive approach to understanding and controlling the dynamic behavior of quadrotors.

OBJECTIVES

- Detailed mathematical modeling of quadrotor systems.
- Linearization of nonlinear equation
- Development of P-PD control algorithm in Simulink.
- Implementation on Real Hardware

METHOD

Quadrotor Dynamics



$$f_t = U_1 = b(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2)$$

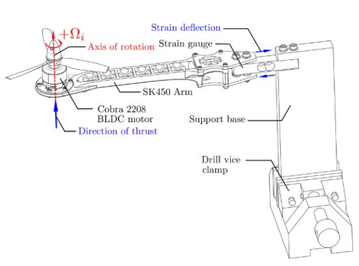
$$T_x = U_2 = bl(-\omega_1^2 - \omega_2^2 + \omega_3^2 + \omega_4^2)$$

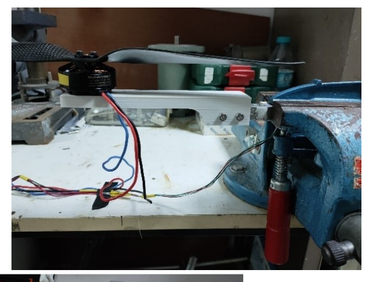
$$T_y = U_3 = bl(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2)$$

$$T_z = U_4 = d(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$$

Images of quadrotor types in the shapes of plus and X.

BLDC Motor System Identification





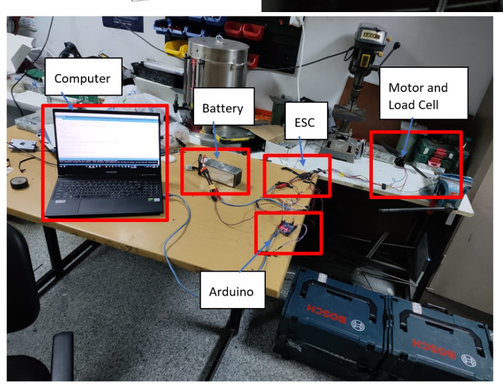
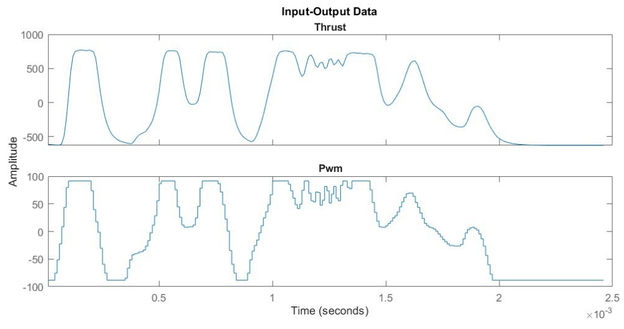
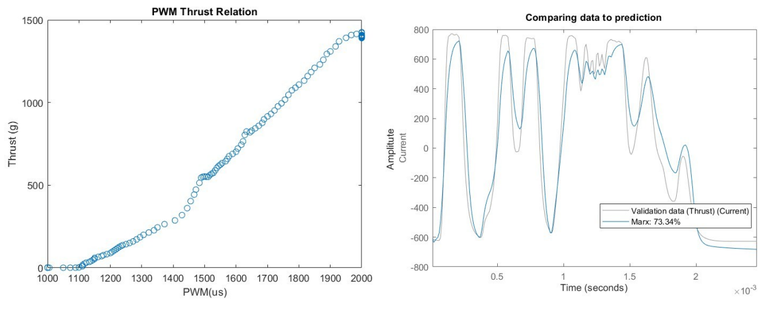
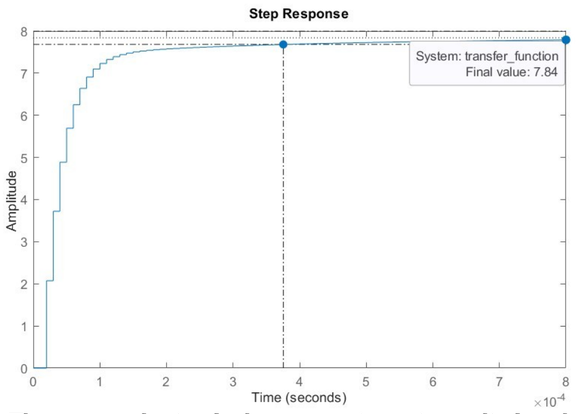


Diagram of the setup created for thrust calculation of BLDC motor.





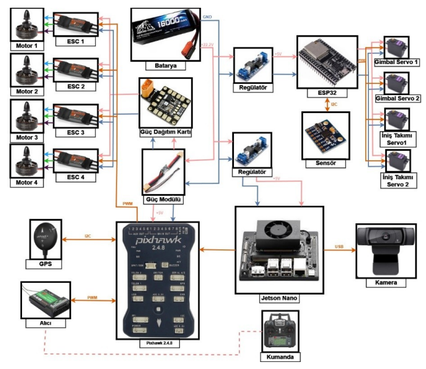
The graph of the dataset obtained from input PWM and output Thrust



The output obtained when a step input is applied to the transfer function of the BLDC motor

MECHANICAL AND ELECTRICAL DESIGN

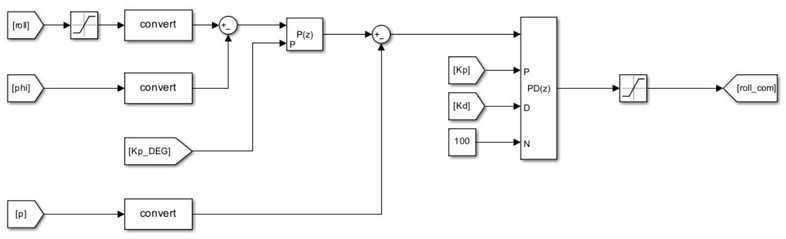


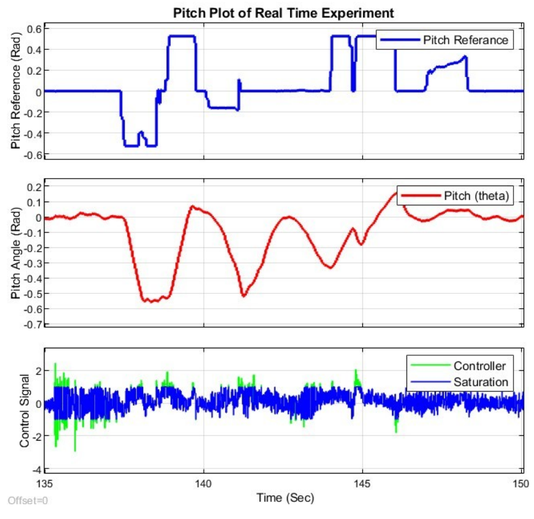


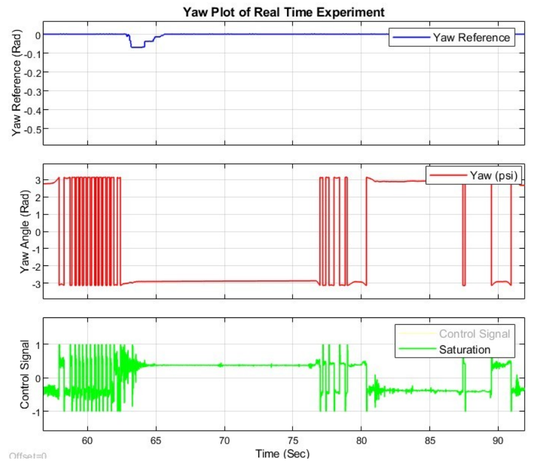
Mechanical and Hardware design of the quadrotor

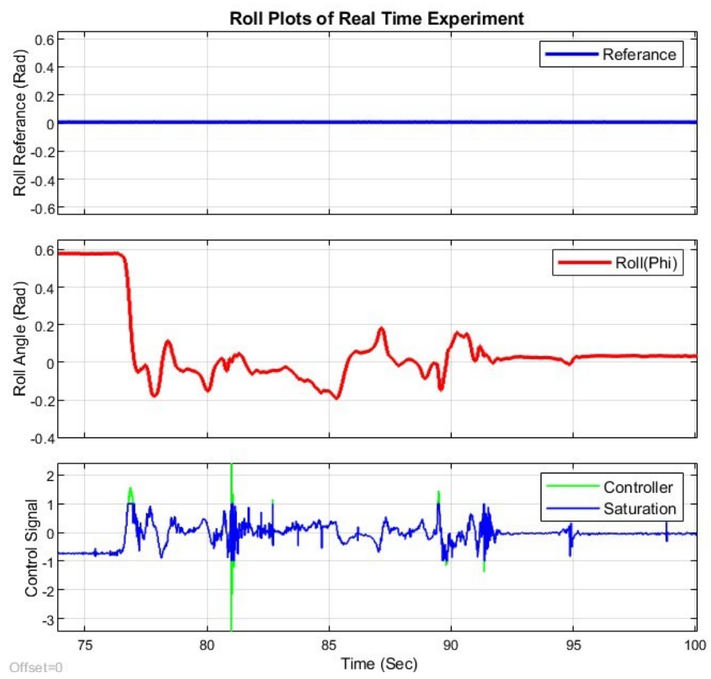
Controller Design (P-PD Controller)

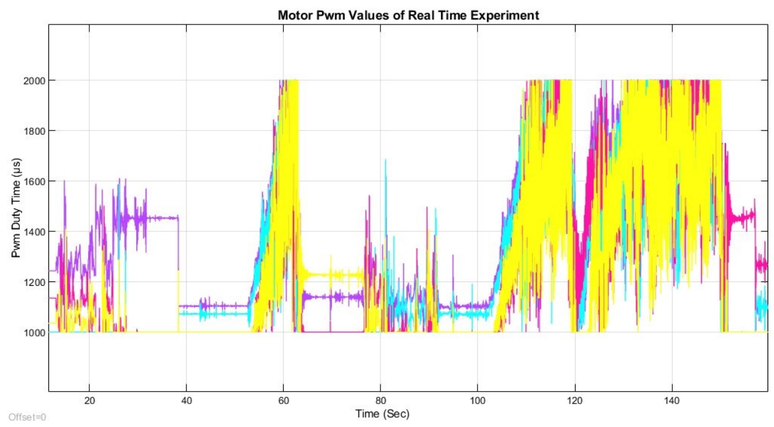
In the development of a robust control system for a quadcopter, Simulink is employed to model and simulate the Proportional (P) and Proportional-Derivative (PD) controllers, integral to the stabilization and control of the quadcopter's roll, pitch, and yaw movements. This academic report outlines the design and implementation of the control model, emphasizing the discrete-time nature of the controllers, data type conversions, and parameter tuning for optimal performance.



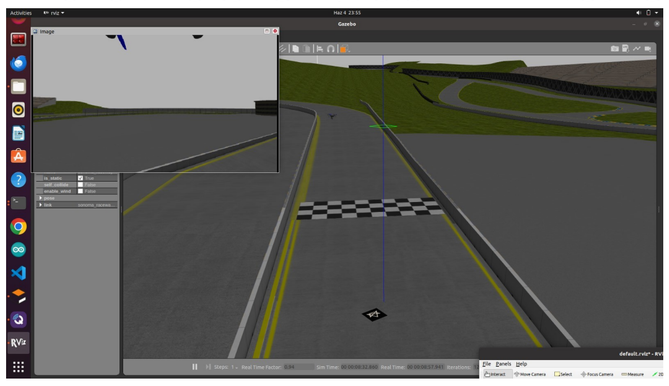


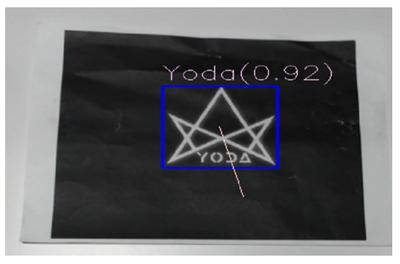






THE FUTURE OBJECTİVES OF THE PROJECT





RESULT

The quadrotor modeling, control, and simulation study achieved significant outcomes. The developed P-PD cascade controller effectively tracked the constant reference signal of 0 radians from an initial orientation. The system's dynamics, especially in yaw motion, were well-represented in the graphs. The PWM duty time graph demonstrated the motor control responses, showing stabilization over time despite its complexity. This confirms the robustness and precision of the control system, validating its efficacy in real-time applications.

REFERENCES

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